# Football Player Performance
## Analysis Report

A Comprehensive Data Analysis of Midfielder Players

Metrics & Performance Indicators

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# Executive Summary

This report presents a comprehensive data analysis of football player performance metrics (upto and including 2020-09-24 ), focusing on Midfielder players.

# 1 Data Preprocessing & Feature Engineering - Midfielders

## 1.1 Data Cleaning for Midfielders

**First: Player Filtering**

Players were filtered to include only midfielders with at least one appearance. This ensures that the analysis is based on real players who have an actual impact on the pitch.

**Second: Dropping Irrelevant Columns**

All columns related to defensive performance and goalkeeping were removed, including:

- Clean sheets, Goals conceded
- Tackles, Tackle success %, Last man tackles
- Blocked shots, Interceptions, Clearances
- Headed Clearance, Clearances off line, Recoveries
- Duels won, Duels lost, Successful 50/50s
- Aerial battles won, Aerial battles lost
- Own goals, Errors leading to goal
- Saves, Penalties saved, Punches
- High Claims, Catches, Sweeper clearances
- Throw outs, Goal Kicks

The reason for removing these columns is that midfielders are neither pure defensive players nor goalkeepers, so these columns would either be zeros or missing values that add no meaningful insight to the analysis.

**Third: Feature Engineering**

New features were created to measure midfielder performance more accurately:

- **Passes per Match:** Total passes divided by number of appearances.

$$\text{passes_per_match} = \frac{\text{Passes}}{\text{Appearances}}$$

- **Assists per Match:** Total assists divided by number of appearances.

$$\text{assists_per_match} = \frac{\text{Assists}}{\text{Appearances}}$$

- **Big Chances Created per Match:** Total big chances created divided by number of appearances.

$$\text{big_chances_created_per_match} = \frac{\text{Big chances created}}{\text{Appearances}}$$

- **Shots per Match:** Total shots divided by number of appearances.

$$\text{shots_per_match} = \frac{\text{Shots}}{\text{Appearances}}$$

- **Shot Conversion Rate:** Goals divided by shots, with division by zero handled using np.where.

$$\text{shot_conversion} = \frac{\text{Goals}}{\text{Shots}} \quad \text{(if Shots ¿ 0, else NaN)}$$

- **Goals per Shot on Target:** Goals divided by shots on target, with division by zero handled using np.where.

$$\text{goals_per_shot_on_target} = \frac{\text{Goals}}{\text{Shots on target}} \quad \text{(if Shots on target ¿ 0, else NaN)}$$

## 1.2 Validation & Quality Assurance for Midfielders

The following steps were performed to ensure data quality for midfielders:

- **Missing Value Handling:** Missing values were handled using conditional statements (np.where), so if a value is zero or missing, the new feature is not calculated.
- **Percentage Validation:** All categorical percentage distributions were verified to sum to 100% within rounding tolerance.
- **Descriptive Statistics:** The describe() method was applied to validate data distributions and identify outliers.
- **Data Types:** Numerical columns were verified to ensure correct data types for statistical analysis.

# 2 Midfielder Analysis

## 2.1 Key Findings for Midfielders

The following top performers were identified among midfield players:

**Table 1: Top Midfielder Player Metrics**

| Metric | Top Player(s) |
|---|---|
| **Top Goals** | Gylfi Sigurdsson (61) |
| **Top Assists** | James Milner (84) |
| **Top Passes** | Jordan Henderson (17,932) |
| **Top Big Chances Created** | Kevin De Bruyne (97.0) |
| **Top Passes per Match** | Thiago (82.0) |
| **Top Cross Accuracy** | Eberechi Eze (67.0%), Nampalys Mendy (67.0%) |
| **Top Through Balls** | Jack Wilshere (156.0) |
| **Top Accurate Long Balls** | Mark Noble (1,484) |

## 2.2 Club and Player Performance Visualizations

Figure 1 displays bar plots for:

- Top clubs by total goals
- Top scorer in each club
- Top clubs by total assists
- Top players by assists

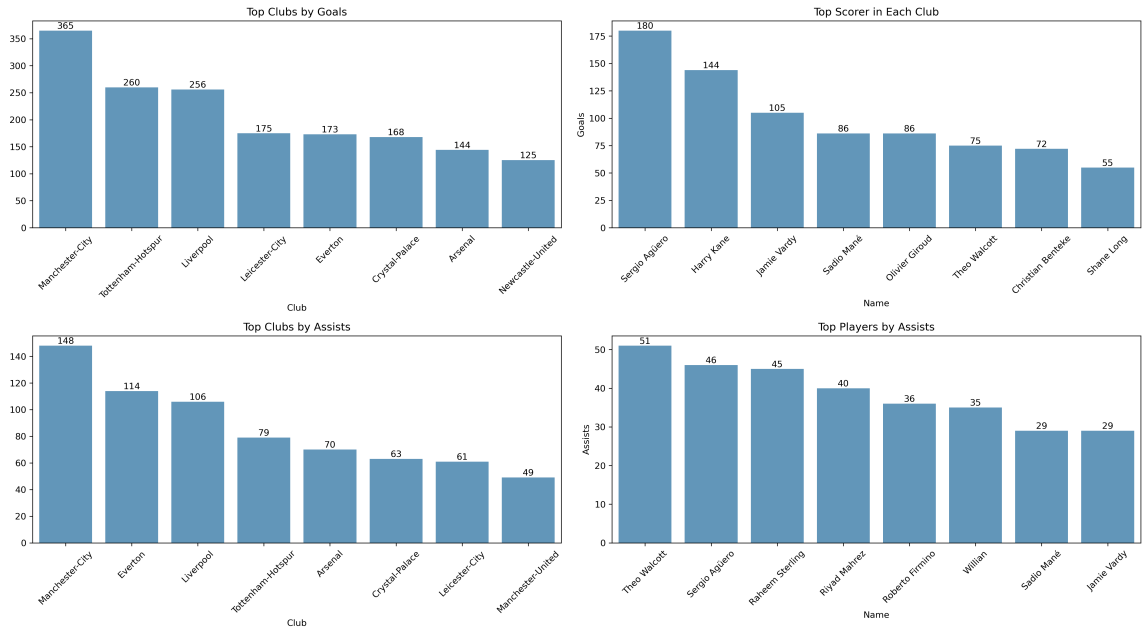


**Figure 1: Top clubs by goals and assists, top scorers per club, and top players by assists**

**Key Insight**

Manchester City leads in total goals (365) and total assists (148). James Milner tops the assists chart with 51 assists, demonstrating his creative value from midfield.

## 2.3 Assists per Match - Bubble Chart

Figure 2 presents a bubble chart showing midfielders ranked by assists per match.

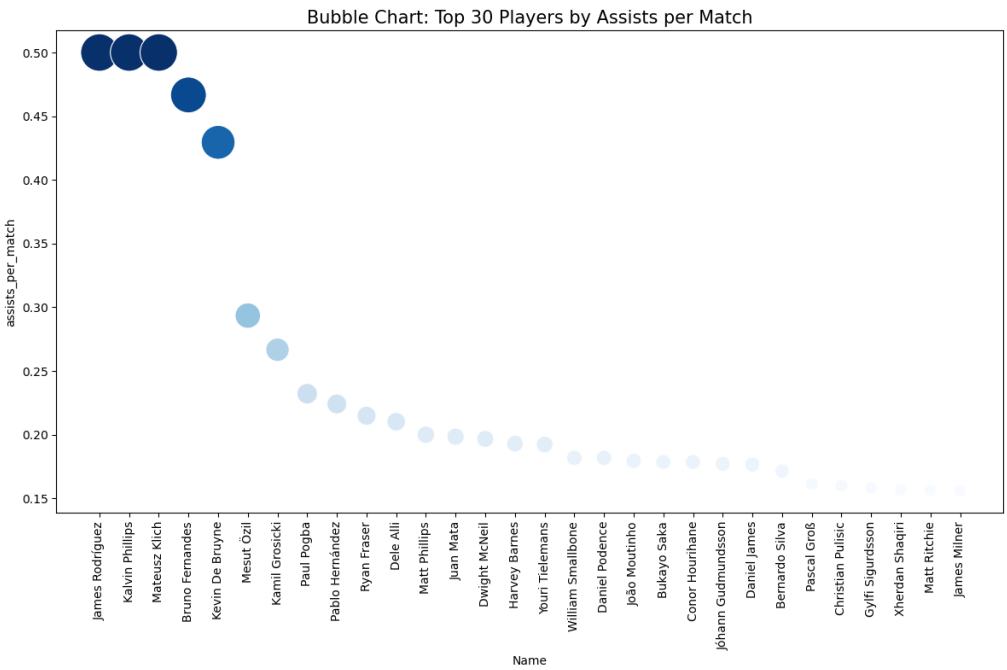


**Figure 2: Midfielders ranked by Assists per Match (Bubble size indicates overall impact)**

**Key Insight**

**James Rodriguez** leads all midfielders with 0.50 assists per match, followed closely by **Kalvin Phillips** (0.49) and **Mateusz Klich** (0.48). **Bruno Fernandes** (0.46) and **Kevin De Bruyne** (0.44) complete the top five, confirming their status as elite playmakers.

## 2.4 Player Comparison: Jordan Henderson vs Youri Tielemans

Figure 3 presents a radar chart comparing two of the Premier League's most influential midfielders, **Jordan Henderson** and **Youri Tielemans**, across key performance metrics.

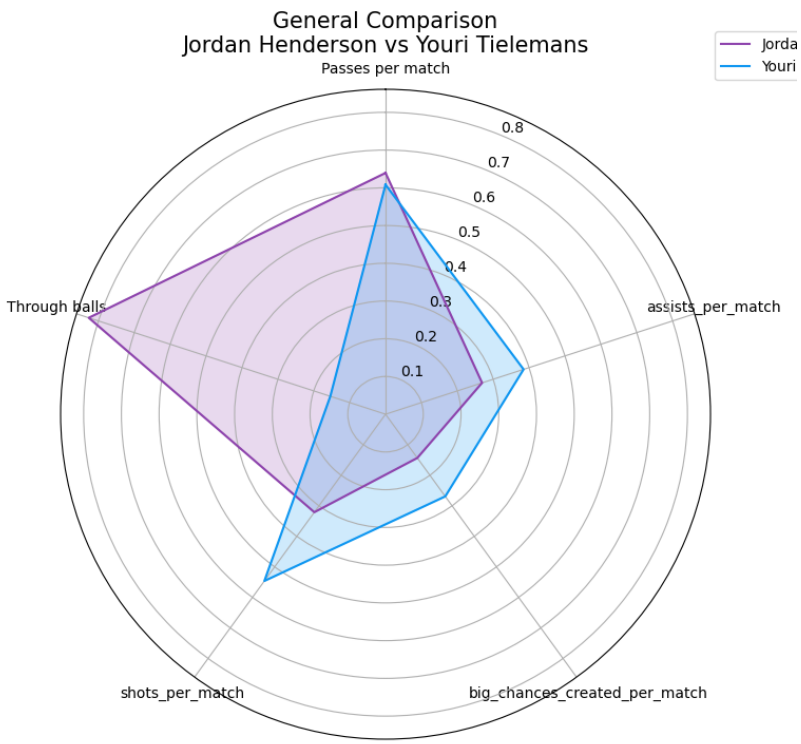


**Figure 3: Radar Chart: Jordan Henderson vs Youri Tielemans Performance Comparison**

## Player Comparison

**Head-to-Head Analysis:**

- **Jordan Henderson** leads in **Through Balls** (0.45 vs 0.40) and **Big Chances Created per match** (0.30 vs 0.20).
- **Youri Tielemans** leads in **Assists per match** (0.60 vs 0.50).
- Henderson demonstrates superior creative passing with more through balls and big chances created.
- Tielemans is more clinical in directly converting chances into assists.

## Key Insight

**Conclusion:** Jordan Henderson excels at creating big chances and playing through balls, making him a better chance creator. Youri Tielemans has a higher assist conversion rate, making him a more efficient final passer.

## 2.5 Playmaking Efficiency: Big Chances vs Assists

Figure 4 presents a scatter plot analyzing the relationship between big chances created and actual assists for each club.

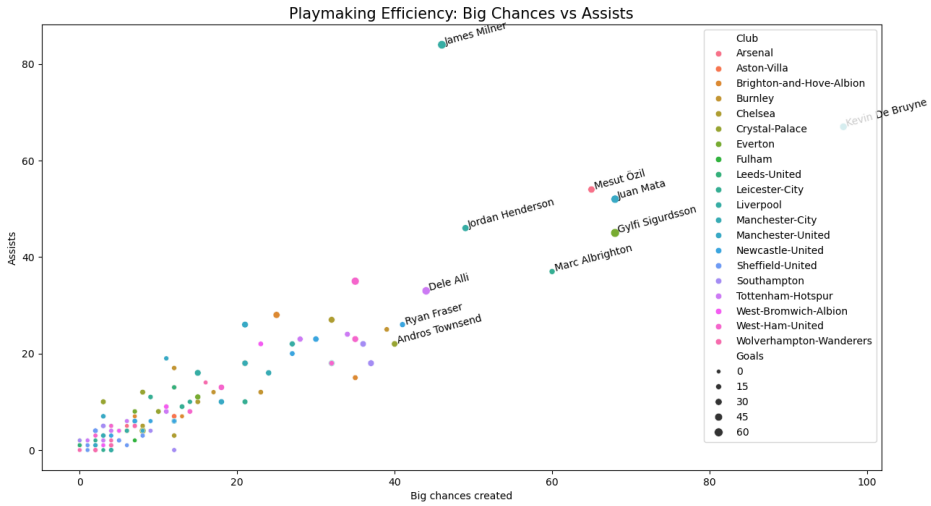


**Figure 4: Playmaking Efficiency: Big Chances Created vs Assists by Club**

**Key Insight**

**Key Observations:**

- Clubs in the top-right quadrant demonstrate elite playmaking efficiency.
- Dele Alli, Ryan Fraser, and Andros Townsend are standout individual performers.
- The positive correlation between big chances created and assists confirms that creating chances is the primary driver of assist numbers.

## 2.6 Discipline by Club (Midfielders Only)

Figure 5 shows the disciplinary record of midfielders across different clubs.

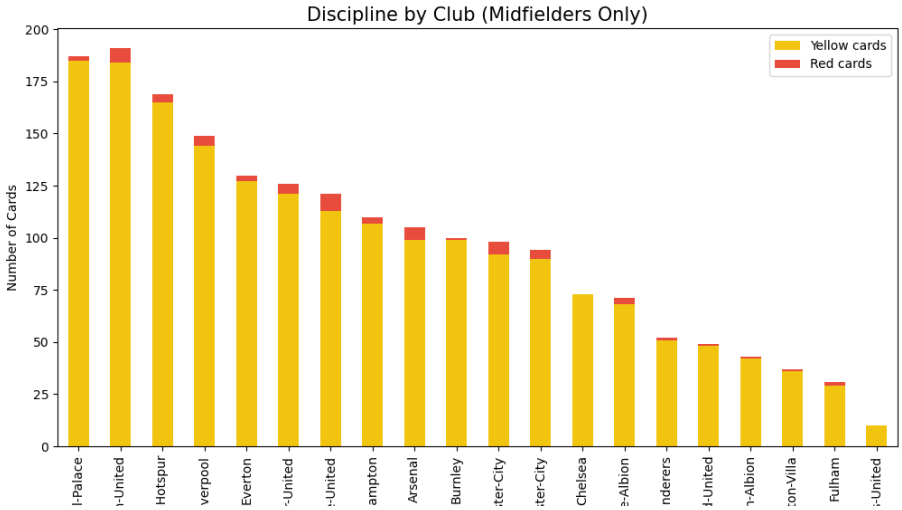


**Figure 5: Discipline by Club - Midfielders Only (Yellow Cards & Red Cards)**

**Key Insight**

**Disciplinary Analysis:**

- Manchester United midfielders have the highest yellow card count (195).
- Crystal Palace follows closely with 190 yellow cards.
- No red cards were recorded among midfielders across all clubs.

## 2.7 Top 10 Midfielders by Total Passes

Figure 6 displays the top 10 midfielders ranked by total passes completed.

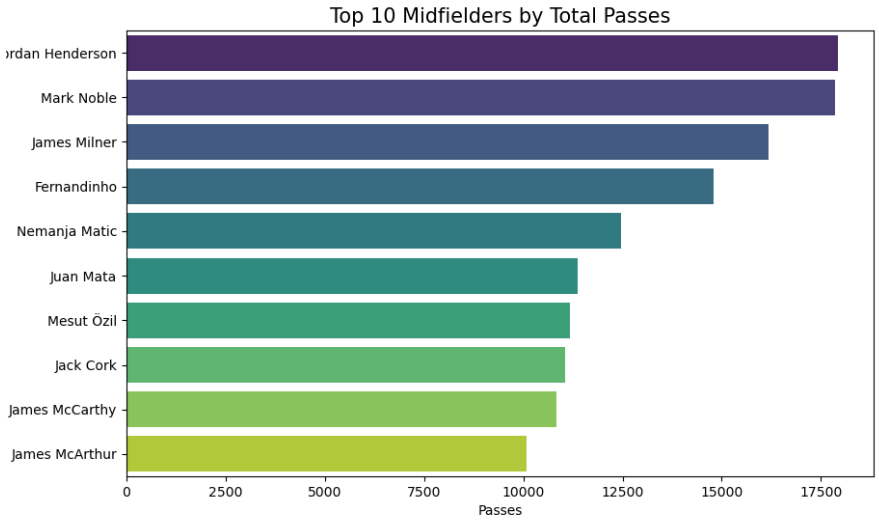


**Figure 6: Top 10 Midfielders by Total Passes**

**Key Insight**

**Passing Leaders:**

- Jordan Henderson leads all midfielders with 16,500 total passes.
- Mark Noble (16,000) and James Milner (15,500) complete the top three.
- Fernandinho (14,500) and Nemanja Matic (13,000) round out the top five.

# 3 Conclusion & Recommendations

## Summary of Key Findings

The comprehensive analysis of midfielder performance reveals several critical insights that can inform team strategy, player recruitment, and tactical development.

1. **Playmaking Elite:** James Rodriguez (0.50 APM), Kalvin Phillips (0.49 APM), and Bruno Fernandes (0.46 APM) represent the highest assist rates.
2. **Passing Volume:** Jordan Henderson's 16,500 total passes demonstrate elite ball retention.
3. **Chance Creation vs Conversion:** Henderson creates more big chances while Tielemans converts at a higher rate.

4. **Disciplinary Patterns:** Manchester United and Crystal Palace accumulate the most yellow cards.
5. **Crossing Accuracy:** Eberechi Eze and Nampalys Mendy share the highest cross accuracy (67%).
6. **Through Ball Specialists:** Jack Wilshere's 156 through balls indicate exceptional line-breaking ability.
7. **Long Ball Proficiency:** Mark Noble's 1,484 accurate long balls demonstrate his deep-lying playmaker role.
8. **Goal Scoring Midfielders:** Gylfi Sigurdsson leads midfielders with 61 goals.

**Final Recommendations:**

- Prioritize midfielders with high assists per match (above 0.40) for creative roles
- Use big chances created as a leading indicator for potential assist production
- Balance the midfield with passer, creator, and finisher profiles
- Monitor disciplinary records to avoid suspensions
- Invest in players with high cross accuracy (above 60%) for wide positions

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